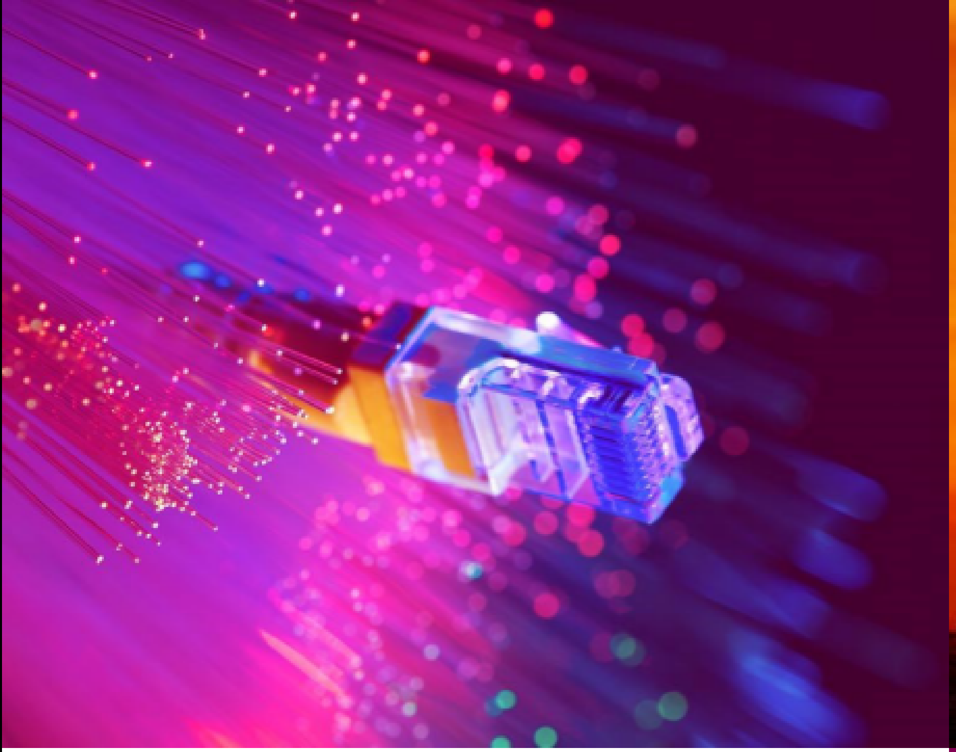

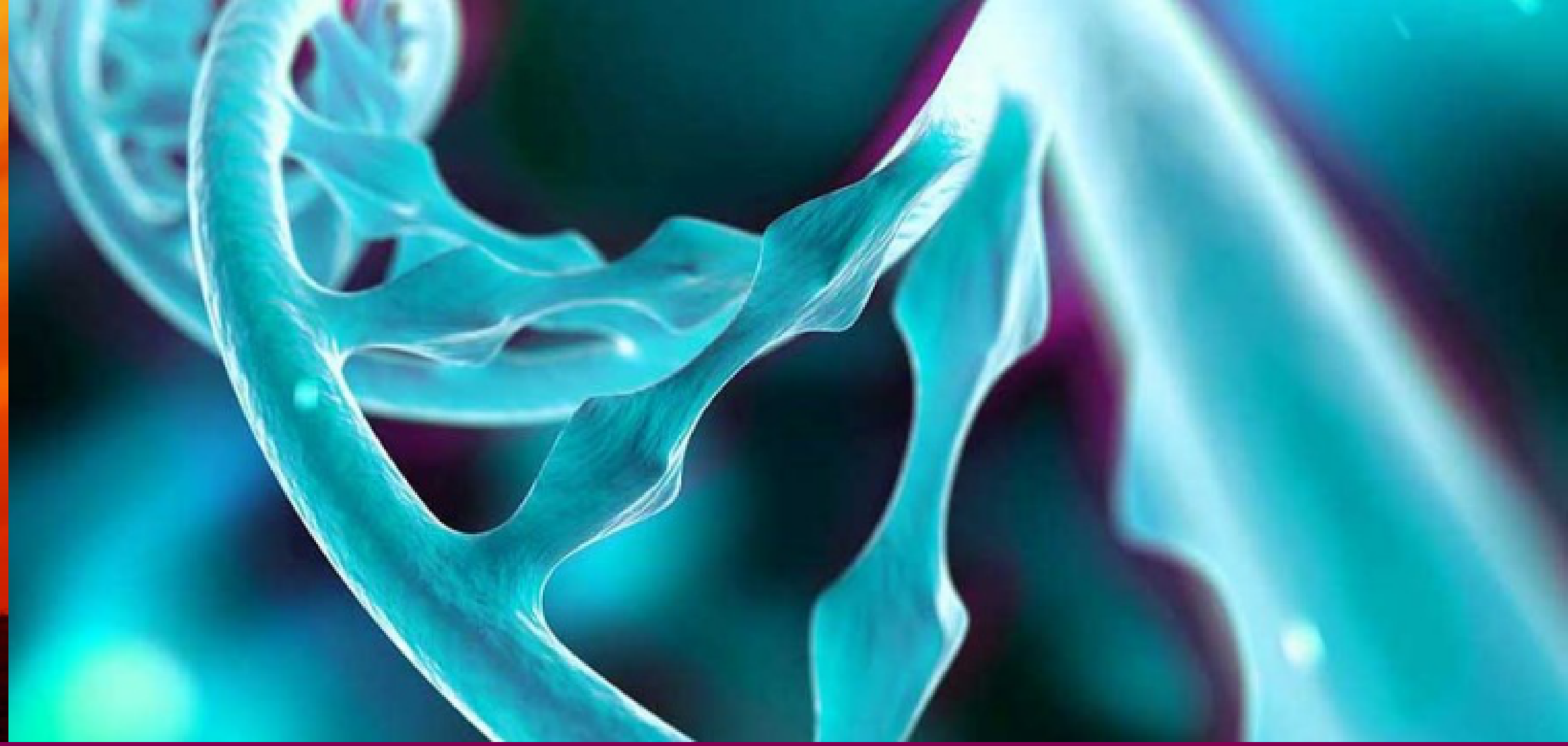

Working in supercomputers

Student cluster competition


CSC – Finnish expertise in ICT for research, education and public administration

Outline

- Module system
- Running applications via batch job system

Some information provided here is specific to LUMI, Mahti, and Puhti, some more general

Modules

Module environment

- Supercomputers have a large number of users with different needs for development environments and applications
- *Environment modules* provide a convenient way to dynamically change the user’s environment
- In this way, different compiler suites and application versions can be used more easily
    - Changing compiler module loads automatically also correct versions of libraries
    - Loading a module for application sets up the correct environment with single command

Common module commands

`module load mod`

Load module **mod** in shell environment

`module unload mod`

Remove module **mod** from environment

`module list`

List loaded modules

`module avail`

List all available modules

`module spider mod`

Search for module **mod**

`module show mod`

Get information about module **mod**

`module switch mod1 mod2`

Switch loaded **mod1** to **mod2**

Batch queue system

Batch queue system

- On a cluster, instead of running a program instantly, you submit your program/simulation (aka job) to a queue and the system will then execute it once the resources are available
  - The queue enables effective and fair resource usage
  - CSC uses SLURM as the queue system
- When running a job on a supercomputer you need to:
  - Describe how you want to run the job and what resources you need
  - Add a command that launches your program
  - Submit your job to a queue
- This is done with a batch job script

Example SLURM batch job script

```
#!/bin/bash
#SBATCH --job-name=example
#SBATCH --account=<project_id>
#SBATCH --partition=debug
#SBATCH --time=00:10:00
#SBATCH --nodes=2
#SBATCH --ntasks-per-nodes=128

# srun launches "nodes * ntasks-per-nodes" copies of myprog
srun myprog
```

- More examples:
  - https://docs.lumi-supercomputer.eu/runjobs/scheduled-jobs/batch-job/
  - https://docs.csc.fi/computing/running/example-job-scripts-puhti/
  - https://docs.csc.fi/computing/running/example-job-scripts-mahti/

Running batch jobs under SLURM

- Submit your batch job script to the queue using `sbatch`

```
sbatch my_job.sh
```

- You can follow the status of your jobs with `squeue`:

```
squeue -u my_username
```

- If something goes wrong, you can cancel your job with `scancel`:

```
scancel jobid
```

  (here the jobid is the numeric ID of the job)

- Show job resource usage (for completed jobs) with `sacct`:

```
sacct jobid
```

Useful Slurm environment variables

Following variables are available inside Slurm scripts:

- `SLURM_JOB_NAME` : name given in `job_name`
- `SLURM_JOBID`
- `SLURM_JOB_NODELIST` : list of nodes the job will run

Following variables are available inside program launched by `srun`

- `SLURM_PROCID` : global id of process
- `SLURM_LOCALID` : node local id of process

Following variables override the corresponding variables in the batch script

- `SBATCH_ACCOUNT` : `--account`
- `SBATCH_PARTITION` : `--partition`